

NITIN BAHUNADAM

Generative AI | Applied AI | Machine Learning Engineer

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PROFESSIONAL SUMMARY

Generative AI, Applied AI, and Machine Learning Engineer with 6+ years of experience developing enterprise AI applications, Retrieval-Augmented Generation systems, NLP pipelines, machine-learning solutions, semantic-search services, data-processing workflows, and secure backend integrations.

Hands-on experience building production-oriented Generative AI applications using Python, FastAPI, LangChain, GPT-4, Vertex AI, Pinecone, FAISS, embeddings, semantic retrieval, prompt engineering, structured outputs, source citation, evaluation workflows, guardrails, PII validation, fallback handling, and operational monitoring.

At American Express, contributed to an Internal Generative AI Knowledge Assistant that enabled authorized employees to retrieve source-grounded answers from approved enterprise documents. Developed document-ingestion, chunking, embedding, retrieval, prompt-orchestration, response-validation, evaluation, and Java-to-Python integration components.

At ADP, developed machine-learning, NLP, data-engineering, semantic-retrieval, and early Generative AI solutions for payroll, HR, workforce analytics, employee attrition, payroll-anomaly detection, policy retrieval, benefits explanations, and employee-support workflows. Integrated Python and FastAPI-based model services with enterprise Java and Spring Boot applications.

Earlier at LTIMindtree, established a strong enterprise-engineering foundation by developing Java and Spring Boot microservices, REST APIs, Angular and React applications, event-driven integrations, database solutions, automated tests, cloud deployments, and limited ML-model integrations for healthcare and insurance platforms.

Experienced across the AI-delivery lifecycle, including use-case discovery, data preparation, document ingestion, model development, RAG architecture, prompt design, model serving, evaluation, explainability, security, deployment, monitoring, production support, and stakeholder collaboration.

CORE COMPETENCIES

Generative AI | Applied AI | Machine Learning | Natural Language Processing | Retrieval-Augmented Generation | Large Language Models | Semantic Search | Vector Databases | Prompt Engineering | LLM Evaluation | AI Guardrails | Model Serving | Java-Python Integration | Data Engineering | Explainable AI | MLOps | Cloud AI | Enterprise AI Integration

TECHNICAL SKILLS

Generative AI and LLMs: Retrieval-Augmented Generation, LangChain, LangGraph, LlamaIndex, OpenAI API, GPT-4, GPT-3, T5, Gemini API, Azure OpenAI, AWS Bedrock, Claude API, Prompt Engineering, Context Engineering, Structured Outputs, Tool Calling, Source Citation, Guardrails, Evaluation Workflows, Multi-Agent Orchestration

Retrieval and Vector Search: Pinecone, FAISS, Chroma, Weaviate, Qdrant, Redis Vector Search, Elasticsearch, Embeddings, Semantic Search, Hybrid Search, Metadata Filtering, Recursive Chunking, Semantic Chunking, Document Ranking, Retrieval Evaluation

Machine Learning and NLP: BERT, SpaCy, scikit-learn, XGBoost, LightGBM, TensorFlow, Classification, Named-Entity Recognition, Feature Engineering, Anomaly Detection, Attrition Modeling, Model Evaluation, Error Analysis, SHAP, LIME

Programming Languages: Python, Java 17, SQL, PL/SQL, TypeScript, JavaScript

AI Application Development: Python, FastAPI, REST APIs, Pydantic Schemas, Request Validation, Error Handling, Async Service Integration, API Contracts, Model-Service Integration

Data Engineering: Pandas, PySpark, Apache Spark, Apache Airflow, ETL Pipelines, Data Cleaning, Data Transformation, Schema Validation, Data Quality, Structured Data, Semi-Structured Data, Feature Pipelines

Backend and Enterprise Integration: Java, Spring Boot, Microservices, REST APIs, GraphQL, Service-to-Service Communication, OAuth2, JWT, Role-Based Access Control, Transaction Workflows, PostgreSQL, Oracle, MongoDB

Messaging and Streaming: Apache Kafka, RabbitMQ, Event-Driven Architecture, Asynchronous Processing, Producers, Consumers, Retry Handling

Cloud Platforms: GCP Vertex AI, BigQuery, Cloud Functions, Cloud Monitoring; AWS S3, Lambda, SageMaker, Bedrock, IAM; Azure Kubernetes Service, Azure OpenAI; Oracle Cloud Infrastructure

MLOps and DevOps: MLflow, Weights & Biases, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD, Prometheus, Grafana, SonarQube

Testing and Quality: pytest, JUnit, Mockito, Unit Testing, Integration Testing, API Testing, Regression Testing, Retrieval Testing, Prompt Evaluation, Failure-Path Testing

Frontend and Visualization: React, Angular, TypeScript, Streamlit, Tableau, Power BI

Collaboration Tools: Git, GitHub, JIRA

PROFESSIONAL EXPERIENCE

American Express | Phoenix, AZ

Feb 2025 - Present

Generative AI Engineer | Enterprise Java and AI Integration

Environment: Python, FastAPI, GPT-4, LangChain, RAG, Pinecone, Vertex AI, BigQuery, Cloud Functions, PostgreSQL, Java 17, Spring Boot, REST APIs, React, Docker, Kubernetes, Jenkins, pytest, JUnit, Mockito

- Contributed to the design and implementation of an Internal Generative AI Knowledge Assistant that enabled authorized employees to ask natural-language questions and retrieve source-grounded answers from approved enterprise documentation.
- Partnered with product, business, engineering, security, and platform teams to translate document-search challenges into prioritized RAG, security, integration, evaluation, and user-experience requirements.
- Participated in defining supported document sources, access restrictions, citation requirements, fallback behavior, answer-quality expectations, and production-readiness criteria.
- Developed Python and FastAPI services responsible for document retrieval, prompt construction, LLM invocation, response parsing, source-reference generation, validation, and structured API responses.
- Built document-ingestion pipelines to extract text from PDFs, Word documents, internal technical guides, policies, operating procedures, and approved enterprise repositories.
- Integrated OCR utilities to process scanned and image-based documents that could not be handled through standard text extraction.
- Cleaned and normalized extracted text by removing repeated headers, broken formatting, unnecessary characters, duplicated content, and other ingestion noise.
- Implemented recursive and semantic document-chunking strategies to create retrieval units that preserved meaningful context while remaining suitable for embedding and LLM processing.
- Defined version-aware metadata schemas containing document identifiers, source names, categories, sections, page information, access classifications, ingestion dates, and version details.
- Generated vector embeddings for document chunks and stored embeddings and metadata in Pinecone to support semantic document discovery.
- Indexed 5,000+ enterprise documents and 2M+ associated embeddings in Pinecone to support semantic retrieval across approved knowledge sources.
- Developed Retrieval-Augmented Generation workflows using LangChain, GPT-4, Vertex AI, Pinecone, and approved internal document sources.
- Implemented query-processing steps that normalized user questions, applied metadata filters, generated query embeddings, retrieved relevant document chunks, and passed grounded context to the LLM.
- Created reusable prompt templates containing system instructions, retrieved passages, user questions, source-grounding rules, required output formats, and insufficient-context instructions.
- Designed structured output contracts for generated answers, source citations, document metadata, validation outcomes, workflow states, and error responses.
- Implemented source citation functionality so employees could verify generated answers against the internal documents used during retrieval.
- Developed rule-based, regular-expression, and named-entity validation services to identify PII, restricted content, unsupported statements, incomplete citations, malformed outputs, and policy violations.
- Implemented controlled fallback responses for weak retrieval, missing document context, low-confidence results, unavailable LLM services, malformed responses, and timeout conditions.

- Established prompt and RAG evaluation workflows using representative questions, expected source documents, reviewer feedback, trace logs, retrieval analysis, and response-quality assessments.
- Evaluated system behavior across retrieval relevance, factual grounding, source accuracy, answer completeness, structured-output compliance, and hallucination risk.
- Improved RAG retrieval accuracy from approximately 78% to 91% through chunk-size tuning, metadata filtering, and prompt refinement, within a broader team retrieval-quality target of 88–93%.
- Reduced hallucinated or unsupported responses by 40% through validation rules, structured outputs, and mandatory source citation.
- Diagnosed irrelevant retrieval results and improved quality through chunk-size tuning, overlap adjustments, metadata filtering, relevance thresholds, top-k retrieval changes, and prompt refinements.
- Reviewed unsupported or incomplete LLM responses and improved system instructions, grounding rules, response templates, validation logic, and escalation behavior.
- Built PostgreSQL-backed services for conversation sessions, answer history, source metadata, user feedback, workflow states, validation results, and troubleshooting information.
- Added feedback-capture mechanisms that allowed authorized users and reviewers to flag weak answers, missing sources, or incorrect retrieval results.
- Developed internal REST API contracts for document ingestion, semantic retrieval, answer generation, feedback submission, validation, and conversation persistence.
- Integrated the Python AI service with existing Java and Spring Boot enterprise services.
- Designed Java-to-Python request and response contracts covering authenticated user context, query payloads, structured responses, citations, errors, timeouts, and fallback states.
- Supported Java services responsible for authentication, authorization, session handling, transaction-related processes, audit context, and enterprise workflow coordination.
- Ensured that unauthorized users could not directly access AI endpoints or restricted document collections.
- Implemented structured error mapping between FastAPI AI services and Spring Boot enterprise services.
- Developed Java and Spring Boot REST APIs supporting user authentication, authorization, transaction-related processes, session management, and internal business workflows.
- Supported OAuth2, JWT, and role-based access-control processes for secure access to enterprise application functions and the Internal Knowledge Assistant.
- Maintained 99.9% API availability while supporting 10K–15K+ authenticated requests per day across authentication, transaction, and knowledge-assistant workflows.
- Developed unit and integration tests with pytest for document ingestion, text processing, chunking, retrieval, prompt generation, structured outputs, validation, source citation, and fallback behavior.
- Used JUnit and Mockito to test selected Java service paths, enterprise API integrations, authentication scenarios, transaction workflows, and Java-to-Python communication.
- Maintained 85%+ unit and integration test coverage across Python and Java services supporting the assistant.
- Added operational telemetry for retrieval latency, LLM latency, token usage, validation results, source availability, Java-to-Python communication failures, and application error paths.
- Reduced Java-to-Python integration latency by 30–40% through contract optimization, caching, and asynchronous processing.
- Used Cloud Monitoring and BigQuery to analyze operational behavior, troubleshoot production issues, evaluate releases, and identify workflow bottlenecks.
- Reduced production incidents by 25% through proactive monitoring dashboards and alerting on retrieval and generation failures.
- Supported Docker containerization, Kubernetes deployment, Jenkins CI/CD pipelines, environment promotion, configuration validation, and post-release monitoring.
- Reduced deployment cycle time by 40–50% by streamlining CI/CD pipelines and environment-promotion steps.
- Worked with security and platform teams on IAM permissions, service-account access, environment boundaries, sensitive-data handling, secure logging, and production deployment controls.

- Collaborated within a cross-functional Agile team of 8–10 engineers spanning Java, AI, QA, platform, and security disciplines.
- Prepared technical documentation, API specifications, RAG architecture diagrams, evaluation findings, stakeholder demonstrations, test scenarios, release criteria, and operational-support guides.

ADP | Hyderabad, India

Jul 2021 - Apr 2023

Software Engineer – Machine Learning, NLP and Generative AI

Environment: Python, BERT, SpaCy, scikit-learn, XGBoost, LightGBM, GPT-3, T5, FAISS, Pinecone, FastAPI, Spark, PySpark, Pandas, Airflow, PostgreSQL, Java, Spring Boot, MLflow, Docker, Jenkins, AWS IAM

- Worked on an enterprise HR, payroll-intelligence, and employee-support platform combining machine learning, NLP, data engineering, semantic retrieval, Generative AI prototypes, and enterprise Java integration.
- Collaborated with HR, payroll, engineering, data-governance, compliance, and platform stakeholders to define analytical use cases, data labels, exception categories, evaluation criteria, model-review requirements, and operational handoff processes.
- Collaborated within a 10-member cross-functional engineering team spanning Java, data, and machine-learning disciplines.
- Built NLP pipelines using Python, BERT, and SpaCy to extract skills, named entities, structured fields, employment attributes, and relevant information from HR documents.
- Achieved 92–95% entity-extraction accuracy using BERT and SpaCy across HR documents and performance reviews.
- Processed performance reviews, employee-feedback content, policy documents, workforce records, and semi-structured HR text.
- Developed text-cleaning and normalization components to address inconsistent formatting, incomplete fields, unnecessary characters, duplicated text, and document-quality issues.
- Applied tokenization, sentence processing, named-entity recognition, contextual language modeling, classification, and structured-output generation.
- Reviewed extraction errors and refined NLP logic, entity patterns, preprocessing steps, and output validation.
- Developed reusable data-cleaning, transformation, validation, and feature-engineering components for payroll, tax, benefits, employee, and workforce datasets.
- Built data-processing pipelines using Python, Pandas, Spark, PySpark, and Airflow.
- Processed 10M+ payroll and HR records through these pipelines, reducing overall ETL execution time by 45%.
- Processed both structured and semi-structured datasets and loaded curated outputs into PostgreSQL.
- Implemented schema validation, type conversion, missing-value handling, duplicate detection, date normalization, categorical normalization, aggregation, and pipeline logging.
- Created Airflow workflows with task dependencies, scheduling, failure handling, retries, and operational monitoring.
- Built Airflow workflows processing 500K+ records per day with automated retry and failure handling.
- Prepared repeatable training and validation datasets for employee attrition, workforce analytics, payroll-anomaly detection, and classification use cases.
- Developed and evaluated machine-learning models using scikit-learn, XGBoost, and LightGBM.
- Engineered features from workforce, payroll, benefits, tax, performance, and employee-related data.
- Performed train-validation splits, hyperparameter experimentation, model comparison, threshold analysis, feature review, and business-focused evaluation.
- Conducted false-positive and false-negative analysis to determine whether model results were suitable for analyst review and downstream use.
- Developed employee-attrition models designed to identify workforce patterns and high-risk turnover indicators for authorized HR analysis.
- Improved employee-attrition prediction accuracy to 89% and reduced payroll-anomaly false positives by 25% through threshold tuning.
- Developed payroll-anomaly detection approaches to identify unusual payment, deduction, tax, benefit, or historical-pattern behavior for additional review.
- Applied SHAP feature attribution and custom model-review artifacts to help auditors, compliance stakeholders, and HR users understand model predictions and risk scores.

- Documented influential features, model limitations, review requirements, potential error cases, and conditions requiring human judgment.
- Built FastAPI endpoints to expose NLP extraction, workforce predictions, anomaly scores, and model outputs to downstream applications.
- Defined request schemas, response formats, validation logic, exception handling, health checks, logging, and operational diagnostics for model-serving APIs.
- Supported the transition of machine-learning solutions from experimentation notebooks into reusable services, repeatable pipelines, testable components, and integration-ready APIs.
- Developed Java and Spring Boot services supporting HR, payroll, benefits, employee-service, and workflow-related operations.
- Integrated Java enterprise services with Python FastAPI model endpoints.
- Defined structured contracts for Java services to submit approved model inputs and receive predictions, entity extractions, classification outputs, or anomaly indicators.
- Implemented Java-side response validation, timeout handling, exception handling, error mapping, logging, and controlled fallback behavior.
- Embedded ML and NLP results into existing employee-service and HR-support workflows.
- Designed semantic-retrieval prototypes using FAISS and Pinecone over approved HR policies, payroll documentation, tax materials, and employee-benefit information.
- Cleaned and divided policy documents into searchable sections, generated embeddings, stored vectors, and evaluated semantic-retrieval results.
- Improved natural-language document lookup by retrieving relevant content based on meaning rather than requiring exact keyword matches.
- Prototyped Generative AI workflows using GPT-3 and T5 for payroll-exception summaries, employee-benefit explanations, HR-policy answers, and structured support content.
- Created reusable prompt templates containing source passages, user questions, required formats, grounding requirements, and uncertainty-handling instructions.
- Developed evaluation datasets containing representative HR and payroll questions, expected document sources, acceptable answer structures, and escalation cases.
- Added source-grounding expectations and automatic escalation behavior for questions that lacked sufficient supporting information.
- Evaluated generated responses for relevance, consistency, clarity, source support, completeness, and unsupported claims.
- Refined prompts and retrieval behavior based on human feedback, error analysis, weak responses, and uncertain-answer cases.
- Implemented PII masking, role-aware access controls, audit logging, and AWS IAM policies to protect sensitive employee and payroll information.
- Worked with governance and compliance stakeholders to document data restrictions, model limitations, approved access, review procedures, and production controls.
- Automated model packaging, experiment tracking, build workflows, and deployment steps using Docker, Jenkins, and MLflow.
- Used MLflow to track experiments, model versions, parameters, evaluation results, and reproducible model builds.
- Supported model-service testing, API troubleshooting, Airflow monitoring, pipeline-failure investigation, deployment validation, and operational documentation.

LTIMindtree | Hyderabad, India

Aug 2018 - Jun 2021

Java Full Stack Developer | Machine Learning Project Contributions

Environment: Java, Spring Boot, Microservices, REST APIs, Angular, React, TypeScript, Kafka, RabbitMQ, Oracle PL/SQL, OAuth2, JWT, Docker, Azure Kubernetes Service, Jenkins, JUnit, Mockito, Elasticsearch, scikit-learn, Tableau, Power BI

- Developed Java and Spring Boot microservices for enterprise healthcare and insurance claim-processing applications.
- Supported healthcare and insurance claim-processing workloads handling 100K+ claims per month.

- Built REST APIs for claim-related workflows, application validation, workflow-status processing, healthcare-information access, and downstream integrations.
- Implemented controller, service, validation, integration, and data-access components using layered Java application architecture.
- Designed and maintained versioned service-to-service and user-facing API contracts.
- Implemented request validation, standardized responses, exception handling, business rules, and integration error management.
- Built Angular and React user-interface components using TypeScript and connected application workflows with Spring Boot APIs.
- Implemented OAuth2, JWT authentication, and role-based access controls for protected enterprise application features.
- Integrated Apache Kafka and RabbitMQ to support asynchronous communication, event-driven processing, retry mechanisms, and decoupled workflows.
- Improved Kafka consumer throughput by 25% through tuning and decoupled workflow design.
- Developed and supported producers and consumers for healthcare and insurance-processing events.
- Tuned Oracle SQL queries, PL/SQL stored procedures, indexes, and execution plans to resolve performance and concurrency bottlenecks.
- Reduced average query response time by 35% through query and index optimization.
- Containerized Java services with Docker and supported deployments to Azure Kubernetes Service.
- Built and maintained Jenkins pipelines that automated application builds, tests, quality checks, packaging, and deployment.
- Reduced deployment time by 40% by automating builds and releases through Jenkins and Docker.
- Wrote JUnit and Mockito tests for Spring Boot controllers, service logic, validation, exception paths, database interactions, and external dependencies.
- Achieved 80%+ JUnit test coverage across backend services.
- Implemented application logging and used Elasticsearch-based centralized search to investigate API failures, service errors, messaging problems, and production incidents.
- Collaborated with business analysts, QA engineers, database teams, operations teams, and client stakeholders to translate healthcare and insurance requirements into incremental software releases.
- Worked within an Agile delivery team of 12 developers, QA engineers, and business analysts.
- Contributed to limited machine-learning initiatives involving claims and patient-related data preparation.
- Prepared and cleaned approved data features for risk-scoring and anomaly-detection prototypes.
- Supported scikit-learn-based model experimentation and basic model-evaluation activities.
- Reviewed prototype outputs with technical and business stakeholders to determine whether results were suitable for workflow integration.
- Integrated selected model outputs into existing Java and Spring Boot services through REST endpoints.
- Implemented structured handling of model responses, integration errors, and model-informed workflow results.
- Supported Docker-based deployment of integrated Java and ML components.
- Created Tableau and Power BI dashboards to display model-informed risk indicators, exceptions, trends, and operational measures.

SELECTED GENERATIVE AI AND MACHINE LEARNING PROJECTS

Documentation RAG Assistant

- Built a document-ingestion and retrieval application supporting PDF and document processing, recursive and semantic chunking, metadata enrichment, embedding generation, vector indexing, and source-referenced answers.
- Developed Python and FastAPI endpoints for document ingestion, semantic retrieval, answer generation, feedback capture, and source inspection.
- Implemented logging and diagnostics for weak retrieval, missing context, incomplete source coverage, malformed outputs, and generation failures.

- Added version-aware metadata to distinguish document revisions and improve source traceability.
- Indexed 50K+ document chunks.
- Achieved 90%+ retrieval precision.
- Maintained an average response time under 2 seconds.

RAG InsightHub

- Developed a retrieval-based knowledge assistant that ingests documents from object storage, generates embeddings, stores vectors, and returns grounded answers with citations.
- Built a FastAPI backend supporting document ingestion, question answering, semantic retrieval, response validation, and feedback collection.
- Developed a React interface for document upload, natural-language questions, answer display, source review, and user feedback.

Text-to-SQL Agent Workflow

- Built an agentic natural-language-to-SQL workflow using schema-aware prompting, SQL generation, query validation, execution controls, error correction, and retry logic.
- Implemented validation checks to reduce unsafe, malformed, or unsupported database queries.
- Developed a Streamlit interface for natural-language questions, generated-query review, result visualization, and feedback-driven prompt refinement.
- Documented common failure patterns involving schema misunderstanding, ambiguous user questions, incorrect joins, and SQL-generation errors.
- Generated SQL with a 92% execution success rate.
- Reduced manual reporting effort by 60%.

Real-Time Market Monitoring System

- Built a prototype multi-agent workflow separating market-data ingestion, anomaly review, signal explanation, and risk checks using simulated or publicly available data.
- Defined agent responsibilities, inter-agent communication, workflow traceability, error-handling paths, and alert explanations.
- Focused on transparent analysis and risk review rather than automated trading execution.

Financial Risk and Fraud Detection Prototype

- Developed a machine-learning prototype for transaction-pattern analysis and anomaly identification using sample or publicly available datasets.
- Performed data cleaning, feature engineering, candidate-model comparison, threshold analysis, and evaluation of anomalous transaction behavior.
- Built document-processing components to extract structured fields from sample financial forms.
- Explored how model outputs and supporting evidence could be presented through an analyst-facing interface.
- Achieved 94% fraud detection precision.
- Reduced false positives by 20%.

EDUCATION

University of Central Missouri | Warrensburg, MO
M.S., Big Data Analytics and Information Technology

May 2023 - May 2025

CERTIFICATIONS

- AWS Certified Developer - Associate
- AWS Certified DevOps Engineer - Professional
- Microsoft Certified: Azure Administrator Associate
- Oracle Cloud Infrastructure AI Foundations Associate
- Google 5-Day Generative AI Intensive